

REPO-BACKED APP SUMMARY

FileTransfer

A browser-based LAN file transfer app built with FastAPI, WebSocket signaling, and WebRTC DataChannels. Multiple PCs join the same room, discover peers, and send files directly browser-to-browser while the server coordinates signaling only.

What It Is

A local-network web app for peer-to-peer file transfer, with an optional Windows tray launcher for packaged desktop use.

Who It's For

Primary persona: **Not found in repo.** Inferred from the README and UI: users on the same LAN who need quick ad hoc PC-to-PC transfer in a browser.

How to Run

- Create and activate a virtual environment.
- Install dependencies with **`pip install -r requirements.txt`**.
- Start the app with **`uvicorn app.main:app --host 0.0.0.0 --port 8000 --reload`**.
- Open **`http://127.0.0.1:8000`**; use the machine's LAN IP for another PC on the same network.

What It Does

- Lets devices join a shared room with a room code and display name.
- Shows online peers and allows choosing a specific transfer target.
- Supports files, folders, and drag-and-drop selection.
- Requires receiver approval before a transfer begins.
- Transfers file data browser-to-browser over an ordered WebRTC DataChannel.
- Stores local history and incomplete incoming chunks in IndexedDB for resume-after-refresh.
- Downloads one file directly or bundles multi-file transfers into a browser-built ZIP.

How It Works

- FastAPI serves `/`, `/static`, and `/health`; the frontend lives in `static/index.html`, `styles.css`, and `app.js`.
- A WebSocket endpoint at `/ws/signaling` keeps in-memory room membership and relays join/leave, offer/answer, ICE, transfer, and resume messages through **RoomHub**.
- Each browser creates an **RTCPeerConnection** per peer plus one ordered **file-transfer** DataChannel used for chat messages, file metadata, and binary chunks.
- Incoming session metadata and chunks are stored in IndexedDB (**sessions**, **chunks**); the browser reassembles files locally and creates ZIP output for multi-file sessions.
- For packaged Windows use, **launcher.py** starts a bundled server, exposes local/LAN URLs, manages startup registration, and runs from the system tray.

Evidence used: README.md, app/main.py, server_entry.py, launcher.py, static/index.html, static/app.js